西方经济学

Part 1 Introduction

Lecture 1C Interest and Discount

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Supplement Readings

(1) M9; S2. [1]

(2) 其他资料：[GM] The Data of Macroeconomics

[1] M 指代马工程教材，S 指代课外阅读材料沈坤荣教程。

学习目标

(1) 理解利息与贴现的概念。

(2) 掌握名义利率和实际利率的关系。

(3) 从贴现角度理解资产定价。

(4) 领会马工程教材精神。

Outline

Interest

P: 本金
r: 年利率
t: 投资年限

① Simple Interest 单利
② Annual Compounding 按年复利
③ 按 n 年复利
④ Continuous Compounding 连续复利

Proposition 1

Let P denote the principal of a fixed-income investment made at the end of year 0. Let r be the constant annual interest rate. By simple interest, the value of investment at the end of year t is given by

$$V_S(t) = P(1 + rt). \quad ①$$

By compound interest, if the interest is to compound annually, then the value of investment at the end of year t is $V_C^1(t) = P(1 + r)^t$. ② If it is to compound n times per year, then the value is $V_C^n(t) = P\left(1 + \frac{r}{n}\right)^{nt}$. ③ If it is compounded continuously,

$$V_C(t) = \lim_{n\to\infty} P\left(1 + \frac{r}{n}\right)^{\frac{n}{r}\cdot rt} = Pe^{rt}. \quad ④$$

e^{rt}

The growth rate of x is defined as $\dfrac{x(t+1) - x(t)}{x(t)}$ if x is discrete and $\dfrac{\dot{x}}{x}$ if x is continuous.

增长率

离散变量

连续变量

Outline

1 Interest

2 Discount

3 Examples

4 马工程教材疑难重点

Discount

现值是指未来某笔或某系列支付款项在当前日期的价值

Present value or **present discounted value** is the value on the present date of a payment or series of payments received on other dates.

现值

(反应了资金的时间价值)

Proposition 2 (Discrete-Time) (离散时间下的现值公式)

Assume the interest rate (r) is constant. Let π_t be a cash flow received at the end of year t. If cash flows are received every year from $t = 1$ on, then the present value of ALL cash flows, $\pi_1, \pi_2, \cdots$, at the end of year 0 is

π_t为t年末收到的现金流

$$V(0) = \frac{\pi_1}{1+r} + \frac{\pi_2}{(1+r)^2} + \cdots$$

所有现金流在第0年末的现值

If $\pi_t = \pi$ is constant, then $V(0) = \frac{\pi}{r}$.

恒定不变

$\frac{\pi_t}{(1+r)^t}$ 代表第t年末现金流π_t折算到现今的价值

$(1+r)^t$ 是折现因子

Discount

Proposition 3 (Continuous-Time)

Assume the interest rate (r) is constant. Let $\pi(t)$ be an instantaneous cash flow received at time t. At time 0, the present value of cash flows, $\{\pi(t) : t \in [0, +\infty)\}$, is

$$V(0) = \int_0^{\infty} \pi(t) e^{-rt} dt.$$

If $\pi(t) = \pi$, $V(0) = \frac{\pi}{r}$.

e^{-rt} 是连续时间的折现因子，对应 $\frac{1}{(1+r)^t}$

Outline

费雪效应

Example 1 (Nominal and Real Interest Rates)

The nominal interest rate (名义利率) is the rate of interest that investors pay to borrow money. The real interest rate (实际利率) is the nominal interest rate corrected for the effects of inflation (剔除通胀的影响). What is the relationship between nominal and real interest rates?

For a continuous-time case, the real interest rate $r(t)$ is equal to the nominal interest rate $i(t)$ minus the inflation rate $\pi(t)$. That is,

在连续时间框架中

$$r(t) = i(t) - \pi(t).$$

实际利率 = 名义利率 − 通胀率.

For a discrete-time case, we have an analog:

在离散时间框架中

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}}, \quad \text{or } r_t \approx i_t - \pi_{t+1},$$

精确公式　　近似公式

where $\pi_{t+1} = P_{t+1}/P_t - 1$ is the inflation rate.

$P(t)$ → $P(t+1)$

1 → $e^{i\cdot 1}$

$\frac{1}{P(t)}$ → $\frac{1}{P(t)} e^{r\cdot 1}$ 货物

$\frac{1}{P(t)} \cdot e^{r\cdot 1} P(t+1) = e^{i\cdot 1}$

$e^{r+\pi} = e^{i}$

P_t → P_{t+1} 物价

$\frac{1}{P_t}$ → $\frac{1}{P_t}(1+r_t)$ 1元购买的商品数量

$\pi_{t+1} = \frac{P_{t+1} - P_t}{P_t} \Rightarrow \frac{1}{P_t}(1+r_t) P_{t+1} = 1 + i_t$

$(1+r_t)(1+\pi_{t+1}) = 1 + i_t$

P_t: t期初的股票价格
d_t: t期初宣告,t期末收到的价格
r_f: 无风险资产的收益率
α: 风险溢价
β: $\frac{1}{1+r_f+\alpha}$

Example 2 (Asset Pricing)

Let P_t be the price of a stock at the beginning of period t; d_t be the dividend announced at the beginning of period t but received at the end of period t; r_f be the rate of return on the riskless asset; and α be the risk premium. Free arbitrage between stocks and riskless asset leads to

卖出 买入 股息 无风险收益率 风险溢价

$$\frac{P^e_{t+1} - P_t + d_t}{P_t} = r_f + \alpha, \quad \text{or} \quad P_t = \beta P^e_{t+1} + \beta d_t,$$

$P_t = \frac{P_{t+1}+d_t}{1+r_f+\alpha}$ 现价

计算股票价格

$P_t = \frac{1}{1+r}P_{t+1} + \frac{1}{1+r}d_t$

$P_{t+1} = \frac{1}{1+r}P_{t+2} + \frac{1}{1+r}d_{t+1}$

where $\beta \triangleq \frac{1}{1+r_f+\alpha}$. Find the solution for P_t.

t t+1 t+2 . . . t+T

$P_t = \beta d_t + \beta^2 d_{t+1} + \dots + \beta^T d_{t+T-1} + \beta^T P_{t+T}$

The solution to the difference equation is

$$P_t = \beta^T P^e_{t+T} + \sum_{i=0}^{T-1} \beta^{i+1} d^e_{t+i} = \lim_{T\to\infty} \beta^T P^e_{t+T} + \sum_{i=0}^{+\infty} \beta^{i+1} d^e_{t+i},$$

where $\sum_{i=0}^{+\infty} \beta^{i+1} d^e_{t+i}$ is the fundamental value of a stock.

股票的当前价格等于其未来所有预期股息的现值之和

Example 3 (Bride Price)

彩礼并非养育成本

彩礼是新娘的价格吗？(The Bride Price: A Hmong Wedding Story)

(1) (貞觀) 十六年六月詔。……競結婚媾。多納貨賄。有如販鬻。……積習成俗。迄今未已。既紊人倫。實虧名教。……自今已後。明加告示。使識嫁娶之序。各合典禮。知朕意焉。其自今年六月禁賣婚。……(顯慶) 四年十月十五日詔。……天下嫁女受財。三品已上之家。不得過絹三百匹。四品五品。不得過二百匹。六品七品。不得過一百匹。八品以下。不得過五十匹。皆充所嫁女貲妝等用。其夫家不得受陪門之財。 ——北宋王溥《唐會要·卷八十三·嫁娶》

(2)《中华苏维埃共和国婚姻条例》（1931 年）第八条：“废除聘金，聘礼及嫁妆。”

(3) 《中华人民共和国民法典》（2020 年）第 1042 条：“禁止借婚姻索取财物。”

(4) 2024 年 1 月 17 日，最高人民法院发布《最高人民法院关于审理涉彩礼纠纷案件适用法律若干问题的规定》。2025 年 2 月 28 日，最高人民法院发布《第二批人民法院涉彩礼纠纷典型案例》。

Outline

疑难重点

(1) 掌握 simple interest, compound interest, continuously compound interest, the rate of growth, present value.

(2) 掌握贴现的计算方法。

(3) 掌握 Nominal and real interest rates 之间的关系。

(4) 用贴现的思想理解资产定价公式。

马工程教材疑难重点

1（E2, p.3）

根据马工程教材观点，应当如何评析西方经济学的科学因素和阶级属性?

2（E2, p.19）

根据马工程教材观点，应当如何评析西方经济学的研究对象?